

Victoria Park Academy Shenzhen

CIE IGCSE Computer Science (0478) — YEAR 9 SEMESTER 1

WEEKS 1-6: DATA REPRESENTATION

WEEK 1: BINARY & DENARY

LESSON 1.1 — INTRODUCTION TO BINARY

Topic: Binary & Denary — Introduction

Duration: 40 minutes

Lesson Type: Theory with physical demonstration

Syllabus Ref: 1.1.1

Learning Objective:

Understand why computers use binary; convert positive denary integers (0-255) to 8-bit binary.

Key Vocabulary:

- binary (二进制 / er jin zhi)
- denary (十进制 / shi jin zhi)
- bit (位 / wei)
- byte (字节 / zi jie)
- place value (位值 / wei zhi)

Lesson Structure (40 Minutes):

0-3 min | DO NOW

- | "How many fingers do humans have? How many fingers do computers have?"
- | Discuss in pairs. Key point: computers only understand ON/OFF (1/0).

3-12 min | DIRECT INSTRUCTION

- | Introduce binary as base-2. Teach 8-bit place values: 128, 64, 32, 16, 8, 4, 2, 1.
- | Physical demo: 8 students hold A4 cards with numbers. Stand up = 1, sit down = 0.
- | Class reads the binary number. Example: 5 = students 4 and 1 stand up → 00000101.

12-25 min | GUIDED PRACTICE

- | Convert together: 5 (00000101), 12 (00001100), 25.
- | Students use mini-whiteboards. Teacher circulates, checks understanding.

25-35 min | INDEPENDENT PRACTICE

- | Worksheet 1A: 5 denary → binary conversions.
- | First 2 with printed place value grid, last 3 without.

35-40 min | PLENARY + HOMEWORK

| Exit ticket: "Convert 45 to binary. Write one reason computers use binary."

| Homework: Convert your age, birth month, and birth day to binary.

| Make flashcards for 8 place values (English + Chinese).

EAL Support:

- Physical demonstration essential — kinaesthetic learners and EAL students grasp "standing up = 1" immediately
- Bilingual card on every desk: 二进制 (binary), 十进制 (denary), 位 (bit)
- Place value grid laminated for every student
- Sentence frame: "To convert denary to binary, I find which place values add up to ____"

Differentiation:

Emerging | Use physical cards, peer support, place value grid always visible

Developing | Standard worksheet, occasional hints

Secure | Extension: "What is the smallest number of bits needed for 500?"

Resources Needed:

- A4 place-value cards (128, 64, 32, 16, 8, 4, 2, 1) — 8 sets
- Mini-whiteboards + markers
- Worksheet 1A (printed)
- Laminated place-value desk mats
- Bilingual keyword cards

LESSON 1.2 — BINARY CONVERSION PRACTICE

Topic: Binary & Denary — Conversion Practice

Duration: 40 minutes

Lesson Type: Practice and fluency

Syllabus Ref: 1.1.1

Learning Objective:

Convert binary to denary; convert denary to binary fluently; understand 8-bit limitation.

Key Vocabulary:

- convert (转换 / zhuan huan)
- maximum (最大值 / zui da zhi)
- minimum (最小值 / zui xiao zhi)
- limitation (限制 / xian zhi)

Lesson Structure (40 Minutes):

0-3 min | DO NOW

| Retrieval quiz: "Binary for 7? Binary for 128? What is the largest number in 8 bits?"

| Answers: 00000111, 10000000, 255

3-12 min | DIRECT INSTRUCTION

| Review denary → binary (division method: divide by 2, record remainders).

| Demo: 200 → 11001000. Show binary → denary (add place values).

12-25 min | GUIDED PRACTICE

| Speed challenge: 10 conversions in 10 minutes. Pairs check each other.

| Teacher circulates with answer key.

25-35 min | INDEPENDENT PRACTICE

| Worksheet 1B: Mixed practice (denary → binary, binary → denary).

| Extension: "What is 256 in binary? Why is this a problem?" (9 bits needed, 8-bit overflow)

35-40 min | PLENARY + HOMEWORK

| "A computer stores age in 8 bits. What is the oldest age possible? Is this enough?"

| Homework: Binary crossword puzzle.

EAL Support:

- Division method: write step-by-step on board, students copy format
- Colour-code: binary answers in blue, denary in green
- Pair strong with emerging for speed challenge

Differentiation:

Emerging | Use place value grid, calculator for division method checks

Developing | Standard worksheet, peer checking

Secure | Extension: "How many bits for 1000? For 1 million?"

Resources Needed:

- Worksheet 1B
- Answer key for speed challenge
- Mini-whiteboards

WORKSHEET 1A — BINARY INTRODUCTION

Name: _____ Class: _____ Date: _____

Learning Objective: Convert positive denary integers (0-255) to 8-bit binary

SECTION A: Use the place value grid (first 2 done as example)

Place values: 128 64 32 16 8 4 2 1

Example 1: Denary 5

0 0 0 0 0 1 0 1 → 00000101

Example 2: Denary 12

0 0 0 0 1 1 0 0 → 00001100

1. 1. Denary 25

128 64 32 16 8 4 2 1

— — — — — — — — → —————

2. 2. Denary 100

128 64 32 16 8 4 2 1

_____ → _____

3. 3. Denary 255

128 64 32 16 8 4 2 1

_____ → _____

SECTION B: Without the place value grid

4. 4. Denary 7 → _____

5. 5. Denary 128 → _____

6. 6. Denary 170 → _____

7. 7. Denary 200 → _____

8. 8. Denary 0 → _____

SECTION C: Challenge

9. 9. What is the largest number you can represent with 8 bits?

Answer: _____ (denary) = _____ (binary)

10. 10. What is the smallest positive number you can represent with 8 bits?

Answer: _____ (denary) = _____ (binary)

Bilingual Support:

binary = 二进制 (er jin zhi)

denary = 十进制 (shi jin zhi)

bit = 位 (wei)

byte = 字节 (zi jie)

place value = 位值 (wei zhi)

WORKSHEET 1B — BINARY CONVERSION PRACTICE (MIXED)

Name: _____ Class: _____ Date: _____

SECTION A: Denary → Binary (show working or use division method)

11. 1. 15 → _____

12. 2. 63 → _____

13. 3. 100 → _____

14. 4. 127 → _____

15. 5. 128 → _____

16. 6. 199 → _____

17. 7. 240 → _____

18. 8. 255 → _____

SECTION B: Binary → Denary (add the place values)

19. 9. 00001111 → _____ + _____ + _____ + _____ = _____

20. 10. 10101010 $\rightarrow$ ____ + ____ + ____ + ____ = ____

21. 11. 11110000 $\rightarrow$ ____ + ____ + ____ + ____ = ____

22. 12. 00000001 $\rightarrow$ ____ = ____

SECTION C: Problem Solving

23. 13. A computer stores student ages in 8 bits. What is the oldest age possible?

24. 14. Is 8 bits enough for all ages? Explain:

25. 15. What is 256 in binary? _____

Why is this a problem for 8-bit storage?

ANSWER KEY — WEEK 1

Worksheet 1A:

26. 1. 00011001 (16+8+1=25)

27. 2. 01100100 (64+32+4=100)

28. 3. 11111111 (128+64+32+16+8+4+2+1=255)

29. 4. 00000111

30. 5. 10000000

31. 6. 10101010 (128+32+8+2=170)

32. 7. 11001000 (128+64+8=200)

33. 8. 00000000

34. 9. 255, 11111111

35. 10. 1, 00000001

Worksheet 1B:

36. 1. 00001111

37. 2. 00111111

38. 3. 01100100

39. 4. 01111111

40. 5. 10000000

41. 6. 11000111 (128+64+4+2+1=199)

42. 7. 11110000 (128+64+32+16=240)

43. 8. 11111111

44. 9. 8+4+2+1 = 15

45. 10. 128+32+8+2 = 170

46. 11. 128+64+32+16 = 240

47. 12. 1 = 1

48. 13. 255

49. 14. No, some people live longer than 255. Need more bits.

50. 15. 100000000 (9 bits). Overflow — cannot fit in 8 bits.

WEEK 2: BINARY ADDITION

LESSON 2.1 — BINARY ADDITION RULES

Topic: Binary Addition — Rules and Carry

Duration: 40 minutes

Lesson Type: Theory with practice

Syllabus Ref: 1.1.2

Learning Objective:

Add two 8-bit binary integers using binary addition rules.

Key Vocabulary:

- addition (加法 / jia fa)
- carry (进位 / jin wei)
- column (列 / lie)
- rule (规则 / gui ze)

Lesson Structure (40 Minutes):

0-3 min | DO NOW

- | Binary bingo: Teacher calls denary (5, 12, 17, 31, 64), students mark binary on grid.
- | Winner reads binary aloud.

3-12 min | DIRECT INSTRUCTION

- | Four rules: $0+0=0$, $0+1=1$, $1+1=10$, $1+1+1=11$.
- | Carry to next column (like denary). Demo on board with colour: carries in red, result in blue.

12-25 min | GUIDED PRACTICE

- | Step-by-step: $00101101 + 00010110$. Class does together.
- | Then: $00001111 + 00000001$ (introduce carry chain).

25-35 min | INDEPENDENT PRACTICE

- | Worksheet 2A: 4 addition problems. Error-spotting: 2 broken examples with common errors.

35-40 min | PLENARY + HOMEWORK

- | "Add $11111111 + 00000001$. What do you notice? Where does the carry go?"
- | Homework: Create 4 binary addition problems. Include one with multiple carries.

EAL Support:

- Colour-code consistently: carries = red, sum = blue
- Grid paper essential for column alignment
- Physical: use counting blocks — one colour for 1s, stack to show " $10 = 1 \text{ carry} + 0$ "
- Sentence frame: "I add column by column from right to left, carrying when needed"

Differentiation:

Emerging | Pre-printed grid with columns, work in pairs

Developing | Standard worksheet, self-check with calculator

Secure | Error-spotting + create own problems

Resources Needed:

- Grid paper for binary addition
- Red and blue pens/markers
- Worksheet 2A
- Counting blocks (optional)

LESSON 2.2 — OVERFLOW & EXAM QUESTIONS

Topic: Overflow & Exam Questions

Duration: 40 minutes

Lesson Type: Theory + exam practice

Syllabus Ref: 1.1.2

Learning Objective:

Understand overflow error; identify when overflow occurs; answer exam-style questions.

Key Vocabulary:

- overflow (溢出 / yì chū)
- error (错误 / cuò wù)
- register (寄存器 / jì cún qì)
- 8-bit limit (8 位限制 / 8 wèi xiàn zhì)

Lesson Structure (40 Minutes):**0-3 min | DO NOW**

| "A computer has 8-bit registers. What is the largest number it can store? Add 1 — what happens?"

3-12 min | DIRECT INSTRUCTION

| Overflow: result too large for available bits. $255 + 1 = 0$ (8-bit).

| Real-world: Pac-Man level 256, aircraft systems, millennium bug.

12-25 min | GUIDED PRACTICE

| Identify overflow: $150+150$, $127+128$, $200+100$. Calculate and state Yes/No.

25-35 min | INDEPENDENT PRACTICE

| Worksheet 2B: Exam questions. "Describe overflow and give an example." (4 marks)

35-40 min | PLENARY + HOMEWORK

| "Why is overflow dangerous in a hospital heart monitor? A bank account?"

| Homework: Find one real-world overflow example.

EAL Support:

- "Overflow" = cup overflowing with water — physical analogy
- Sentence frame: "Overflow occurs when the result ____ is larger than the maximum ____"
- Real-world examples make abstract concept concrete

Differentiation:

Emerging | Calculator check, sentence frames for written answers

Developing | Standard exam questions, mark scheme guidance

Secure | Research recent overflow incidents

Resources Needed:

- Worksheet 2B
- Calculator
- Real-world overflow examples

WORKSHEET 2A — BINARY ADDITION

Name: _____ Class: _____ Date: _____

Rules: $0+0=0$ | $0+1=1$ | $1+1=10$ | $1+1+1=11$

Remember to carry to the next column!

SECTION A: Complete these additions (use the grid)

51.1. 128 64 32 16 8 4 2 1

0 0 0 0 1 1 0 1 (13)

+ 0 0 0 0 1 0 1 0 (10)

52.2. 128 64 32 16 8 4 2 1

0 0 0 1 0 1 1 0 (22)

+ 0 0 0 0 1 1 1 1 (15)

53.3. 128 64 32 16 8 4 2 1

0 0 0 1 1 1 1 1 (31)

+ 0 0 0 0 0 0 0 1 (1)

SECTION B: Error Spotting (find and correct)

54.4. 0 0 0 0 1 1 1 1 (15)

+ 0 0 0 0 1 0 0 0 (8)

0 0 0 1 0 1 1 1 (23) ← Is this correct?

Mistake: _____

Correct answer: _____

55.5. 1 1 1 1 1 1 1 1 (255)

+ 0 0 0 0 0 0 0 1 (1)

1 0 0 0 0 0 0 0 (256) ← Is this correct?

Mistake: _____

Correct answer: _____

SECTION C: Challenge

56.6. Add 100 + 100 in binary. Show your working:

Does this cause overflow in 8 bits? _____ Explain: _____

Bilingual Support:

addition = 加法 (jia fa)

carry = 进位 (jin wei)

overflow = 溢出 (yi chu)

error = 错误 (cuo wu)

WORKSHEET 2B — OVERFLOW & EXAM QUESTIONS

Name: _____ Class: _____ Date: _____

SECTION A: Identify Overflow

For each addition, calculate the result and state whether overflow occurs.

57.1. $150 + 150 =$ _____ Overflow? Yes / No

58.2. $127 + 128 =$ _____ Overflow? Yes / No

59.3. $200 + 100 =$ _____ Overflow? Yes / No

60.4. $50 + 50 =$ _____ Overflow? Yes / No

61.5. $255 + 1 =$ _____ Overflow? Yes / No

SECTION B: Exam-Style Questions

62.6. A computer uses 8-bit registers.

(a) What is the largest positive integer it can store? [1 mark]

(b) Explain what is meant by overflow. [2 marks]

(c) Give an example of an addition that would cause overflow. [1 mark]

63.7. Describe two problems that could occur if a computer system experiences overflow when calculating a bank balance. [4 marks]

SECTION C: Real World

64.8. Research: The game Pac-Man has a "kill screen" at level 256.

Explain how overflow is related to this bug.

ANSWER KEY — WEEK 2

Worksheet 2A:

65.1. 00010111 (23)

66.2. 00100101 (37)

67.3. 00100000 (32)

68.4. INCORRECT. Carry from bit 3 to bit 4 was missed. Correct: 00010111 (23)

69.5. INCORRECT. 9-bit result in 8-bit register. Overflow — correct answer is 0 (with overflow flag)

70.6. $100 + 100 = 200$. Binary: $01100100 + 01100100 = 11001000$ (200). No overflow (result < 255).

Worksheet 2B:

71.1. 300, Yes (overflow, max 255)

72.2. 255, Yes (overflow)

73.3. 300, Yes (overflow)

74.4. 100, No

75.5. 0, Yes (overflow)

76.6. (a) 255

(b) Overflow occurs when a calculation result is too large to fit in the available bits.

(c) Any addition where result > 255 , e.g., $200 + 100$

77.7. Two problems: (1) Incorrect balance displayed, (2) Could show negative number or zero,

(3) Financial loss or incorrect transactions, (4) Loss of customer trust

78.8. Level number stored in 8 bits. $255 = \text{max}$. Level 256 = overflow to 0.

Game tries to draw level 0 which doesn't exist properly, causing garbled screen.